Jacobian Analysis of a Recurrent Transformer Block

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Background: Recurrent Transformers

- **Standard Transformers:** Depth is fixed. Increasing depth = increasing parameters.
- **Recurrent Transformers:** Reuse the same layer weights T times.
- **The Problem:** If we loop a block, how do we ensure the activations don't explode or collapse to zero?

Stability and Jacobians

Consider the update: $Y^{(t+1)} = F(Y^{(t)})$.

- **Local Stability:** Determined by the spectral norm of the Jacobian $\|J_F(Y)\|$.
- If $\|J_F\| < 1 \Rightarrow$ The mapping is a local contraction (stable).
- If $\|J_F\| \gg 1 \Rightarrow$ Small perturbations (noise) grow exponentially $\Rightarrow$ Chaos/Explosion.

Research Goal: Find an explicit upper bound for $\|J_F\|$ for a *realistic* Transformer block (MSA + FFN + Residual + RMSNorm).

Building Blocks: MSA and FFN

We analyze the standard Post-Norm configuration:

Multi-Head Self-Attention:

$$A_h = \text{softmax}\left(\frac{Q_h K_h^\top}{\sqrt{d_h}}\right)$$

$$\text{MSA}(Y) = [A_1 V_1 \parallel \cdots \parallel A_H V_H] W_O$$

Feed-Forward Network:

$$\text{FFN}(Y) = \sigma(YW_1 + b_1)W_2 + b_2$$

RMS Normalization:

$$\mathrm{RMSNorm}(y) = \text{diag}(\gamma)\frac{y}{\|y\|_2}$$

Note: RMSNorm is applied row-wise to the $S \times D$ state matrix.

Formulation of the Architecture

The full update $F(Y)$ used in our analysis:

1. **MSA Sub-block:** $Y' = \mathrm{RMSNorm}(Y + \mathsf{MSA}(Y))$
2. **FFN Sub-block:** $\mathsf{TB}(Y) = \mathrm{RMSNorm}(Y' + \mathsf{FFN}(Y'))$
3. **Global Update (with context $C = Y^{(0)}$ and step ν):**

$$F(Y) = \mathrm{RMSNorm}\Big(Y + \nu\big(C + \mathsf{TB}(Y)\big)\Big)$$

Jacobian of RMSNorm

The stability of the entire block rests on this property of RMSNorm:

Lemma (Tomihari & Karakida, 2025):
For $f(u) = \mathrm{RMSNorm}(u)$, the Jacobian $J_f(u)$ satisfies:

$$\left\| \frac{\partial \mathrm{RMSNorm}(u)}{\partial u} \right\|_2 \leq \frac{\gamma_{\max}}{\|u\|_2}$$

- If the input norm $\|u\|$ is large, the Jacobian becomes very small $(1/\|u\|)$.
- RMSNorm acts as a multiplicative damping factor.

Main Result: Spectral Norm Bound (Theorem 1)

Theorem 1. *Suppose inputs to all RMSNorm layers satisfy row-norm lower bounds:*

- $\|(Y + \mathrm{MSA}(Y))\| \geq r_{\mathrm{in}}$,
- $\|(Y' + \mathrm{FFN})\| \geq r_{\mathrm{out}}$,
- $\|(Y + \nu(C + \mathrm{TB}(Y)))\| \geq r_{\mathrm{U}}$,

where $r_{\mathrm{in}}, r_{\mathrm{out}}, r_U > 0$. *Let* $\gamma_{\max} = \max_j |\gamma_j|$. *Then the Jacobian spectral norm is bounded by:*

$$\|J_F(Y)\|_2 \leq \frac{\gamma_{\max}}{r_U}\left[1 + \nu\,\frac{\gamma_{\max}^2}{r_{\mathrm{out}}\, r_{\mathrm{in}}}\left(1 + \|J_{\mathrm{FFN}}\|\right)\left(1 + \|J_{\mathrm{MSA}}\|\right)\right]$$

Main Result: Asymptotic Stability (Theorem 2)

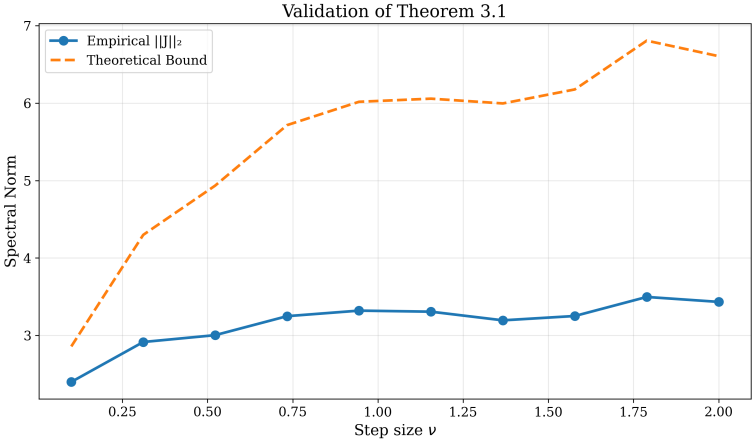
Theorem 2. *Under previous assumptions, and if* $\min_i \left\|\Delta Y_{[i,:]}\right\|_2 \geq \varepsilon > 0$*, the Jacobian norm remains bounded even for infinitely large step sizes* ν*:*

$$\|J_F(Y)\|_2 = O(1) \quad \text{as} \quad \nu \to \infty$$

Experimental Setup

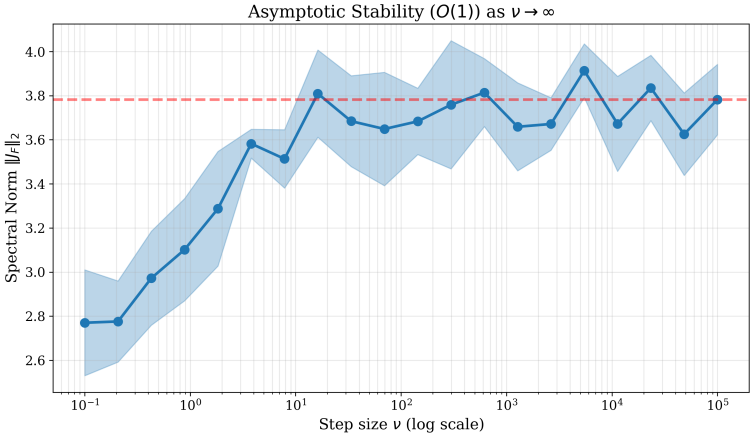
- **Task:** Verify if the theoretical bound is tight and if $O(1)$ holds.
- **Configurations:** $S = 32, D = 64, H = 4, D_{ff} = 128$.
- **Data:** Synthetic Gaussian inputs and CIFAR-10 embeddings.

Experiment 1: Validating Theoretical Bounds



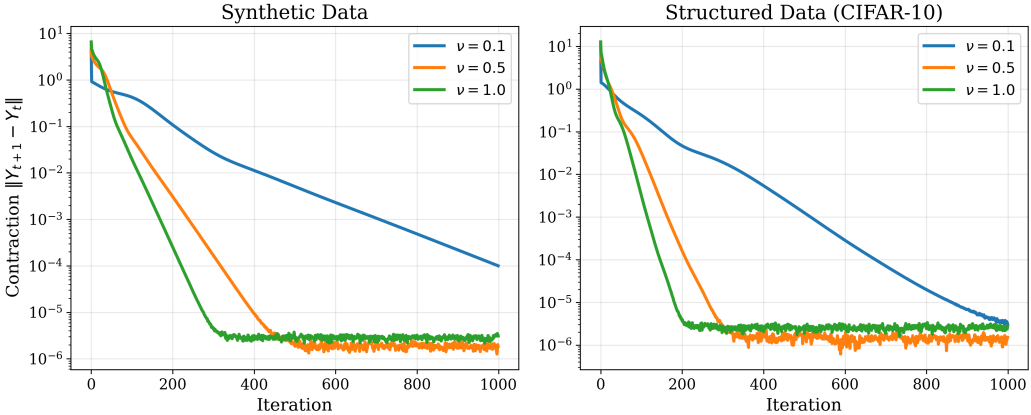
The bound is conservative but tracks the trend of the Jacobian as ν increases.

Experiment 2: Validating Asymptotic Stability



RMSNorm makes the architecture immune to gradient explosion from large step sizes.

Experiment 3: Contraction on Embeddings



Convergence is fast for both synthetic data and real CIFAR-10 embeddings.

Conclusion & Future Work

Contributions:

- Jacobian analysis of a full Transformer block.
- Explicit spectral bounds derived.
- Proved $O(1)$ stability for large ν.

Future Directions:

- Tightening the spectral bounds.
- Extension to Pre-Norm architectures.

Thank you!

Paper, Slides & Code:

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